
ECSE 4660/6660: Internetworking of Things

Lecture 10: Link Layer
Fundamentals – Part II: WiFi,
Cellular

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ECSE, RPI

Administrative

- Use Piazza for any questions
 - <https://piazza.com/rpi/spring2026/e/cse46606660>
- Today's Quiz is available on **LMS**
 - Multiple choice questions
 - Due tonight 11:59 PM
- Lab
 - TA hours 4-5 PM daily in 5312 lab
 - Lab 2 late due Tue, 2/17 *to day*
 - Lab 3 due Fri, 2/27
- HW2 on **LMS** due Mon 2/23 *}*
 - No Late submission allowed
- Project group formation
 - [IoT_SP26_project.xlsx](#)
- Project Next Steps
 - Work with TA to get the hardware needed
 - Mid-term report **due 3/20**
 - In-class presentation+2min video 4/20, 4/23
 - Live-demo 4/27
 - Final code/report 5/1

Link Layer Techniques: Medium Access ^{3 4} (4 Lectures)

- MAC protocols: Aloha, CSMA
 - RFID, LoRa: Aloha, Slotted-Aloha
 - Wired Network: CSMA/CD
 - IEEE 802.11 b/g/n/ac WiFi: CSMA/CA
 - Cellular MAC (NB-IoT): CDMA, TDMA, FDMA, OFDMA
- Other IoT Protocols
 - Bluetooth Low Energy (BLE)
 - IEEE 802.15.4 and Zigbee

LoRa, RFID

5

References

Link Layer Fundamentals: Random Access MAC Protocols

Reference for this part:

- A. Kumar, D. Manjunath, J. Kuri, “Communication Networking, An Analytical Approach,” 2004, Elsevier. (Available as **E-BOOK** through the **RPI Library**)

Chapter 8: Sections:

8.5.2: Carrier Sensing Protocols

8.6.1: The IEEE 802.11 Standard

8.6.2: Multiple Access in IEEE 802.11

- J.F. Kurose, K.W. Ross, “*Computer Networking, A Top-Down Approach Featuring the Internet*” (7th Ed.).

Chapter 7: Wireless Networks (posted on Piazza)

CSMA

References

Reference for Cellular (3G-5G):

V. Garg, Wireless Communications & Networking, Morgan Kauffman, 2007. Available as **E-BOOK** through the RPI Library)

Chapter 6:

- 6.1: Introduction
- 6.2: Narrowband systems
 - 6.2.1 FDD/TDD
 - 6.2.2 FDMA
 - 6.2.3 TDMA
- 6.4 Wideband systems (CDMA)
- 6.5 Comparison between TDMA, FDMA and CDMA

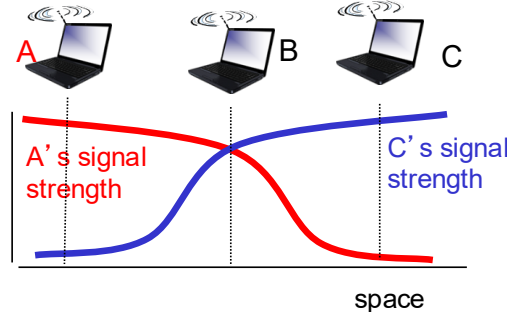
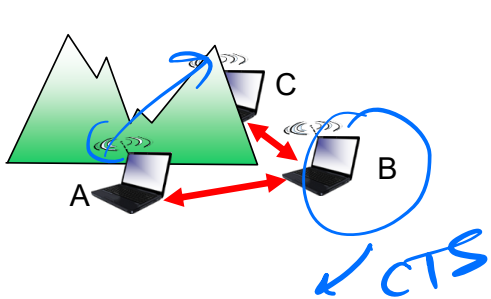
J.F. Kurose, K.W. Ross, "Computer Networking, A Top-Down Approach Featuring the Internet" (7th Ed.).

Chapter 7: Wireless Networks

cellular

WiFi - IEEE 802.11 MAC Protocol: CSMA/CA

- avoid collisions: 2⁺ nodes transmitting at same time
- 802.11: CSMA - sense before transmitting
 - don't collide with detected ongoing transmission by another node
- 802.11: no collision detection (no CD)!
 - difficult to sense collisions: Radios are NOT full-duplex
 - high transmitting signal, weak received signal due to fading
 - can't sense all collisions in any case: hidden terminal, fading
 - goal: *avoid collisions*: CSMA/CollisionAvoidance



Hidden terminal problem!

A and C cannot listen to each other or "hidden" from each other

RTS / CTS

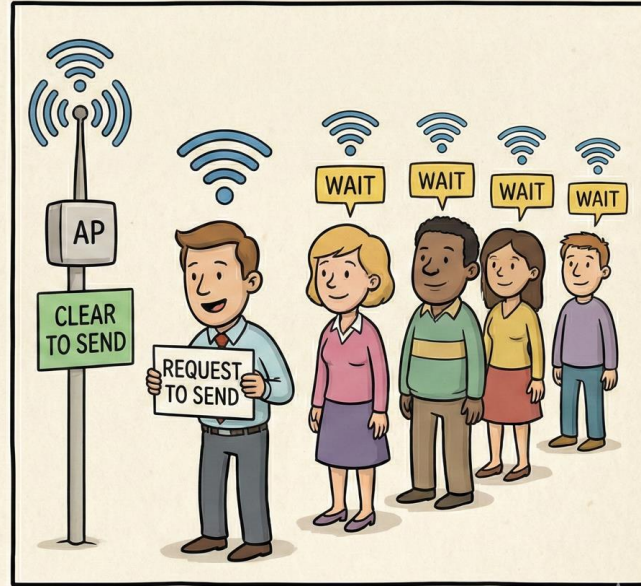
WiFi - IEEE 802.11 MAC Protocol: CSMA/CA

CSMA/CD (Wired): "I'll stop if I hear someone else."



Listen while sending. Polite interruption.

CSMA/CA (Wireless): "I'll wait for my turn to speak."



Listen before sending. Patient waiting.

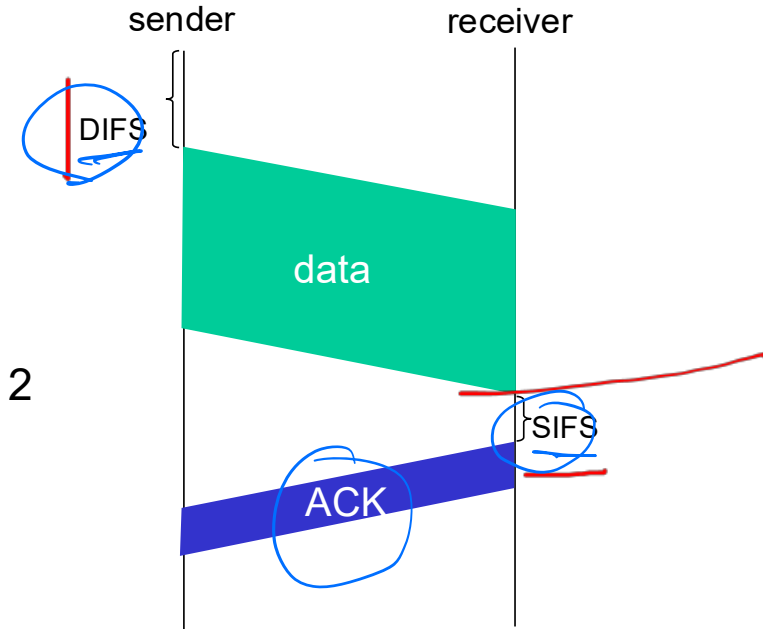
WiFi - IEEE 802.11 MAC Protocol: CSMA/CA

802.11 sender

- 1 if sense channel idle for **DIFS** then
transmit entire frame (no CD)
- 2 if sense channel busy then
 - start random backoff time
 - timer counts down while channel idle
 - transmit when timer expires
 - if no ACK, increase random backoff interval, repeat 2

802.11 receiver

if frame received OK
return ACK after **SIFS** (ACK needed due to hidden terminal problem)



Q: Why SIFS is shorter duration than DIFS?

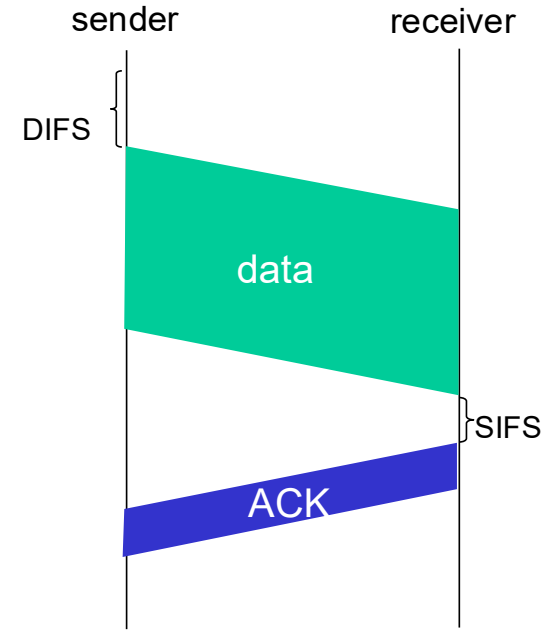
DIFS (Distributed Inter-Frame Space), SIFS (Short Inter-Frame Space)

Events during 802.11 DCF exchange

- Time measured in terms of a basic time-slot (depends on PHYs, e.g., 9 ms in 802.11a)
- IFS
 - Inter-frame spacing
 - Transmission of a pkt is allowed only after appropriate IFS has passed while channel remains idle
 - SIFS: the shortest of all, used for high priority frames such as control messages. Turn around time.
 - DIFS: DCF IFS.
 - SIFS (Short IFS) < DIFS
 - PIFS (used for PCF takeover), SIFS < PIFS < DIFS
 - Other IFS: EIFS, AIFS, RIFS

RTS
CTS
ACK

Shortest



The DIFS duration in 802.11ac can be calculated as: $DIFS = SIFS + 2 \times \text{Slot Time}$

- The Slot Time in 802.11ac is $9 \mu s$
- Therefore: $DIFS = 16 \mu s$ (SIFS) $+ 2 \times 9 \mu s$ (Slot Time) $= 34 \mu s$

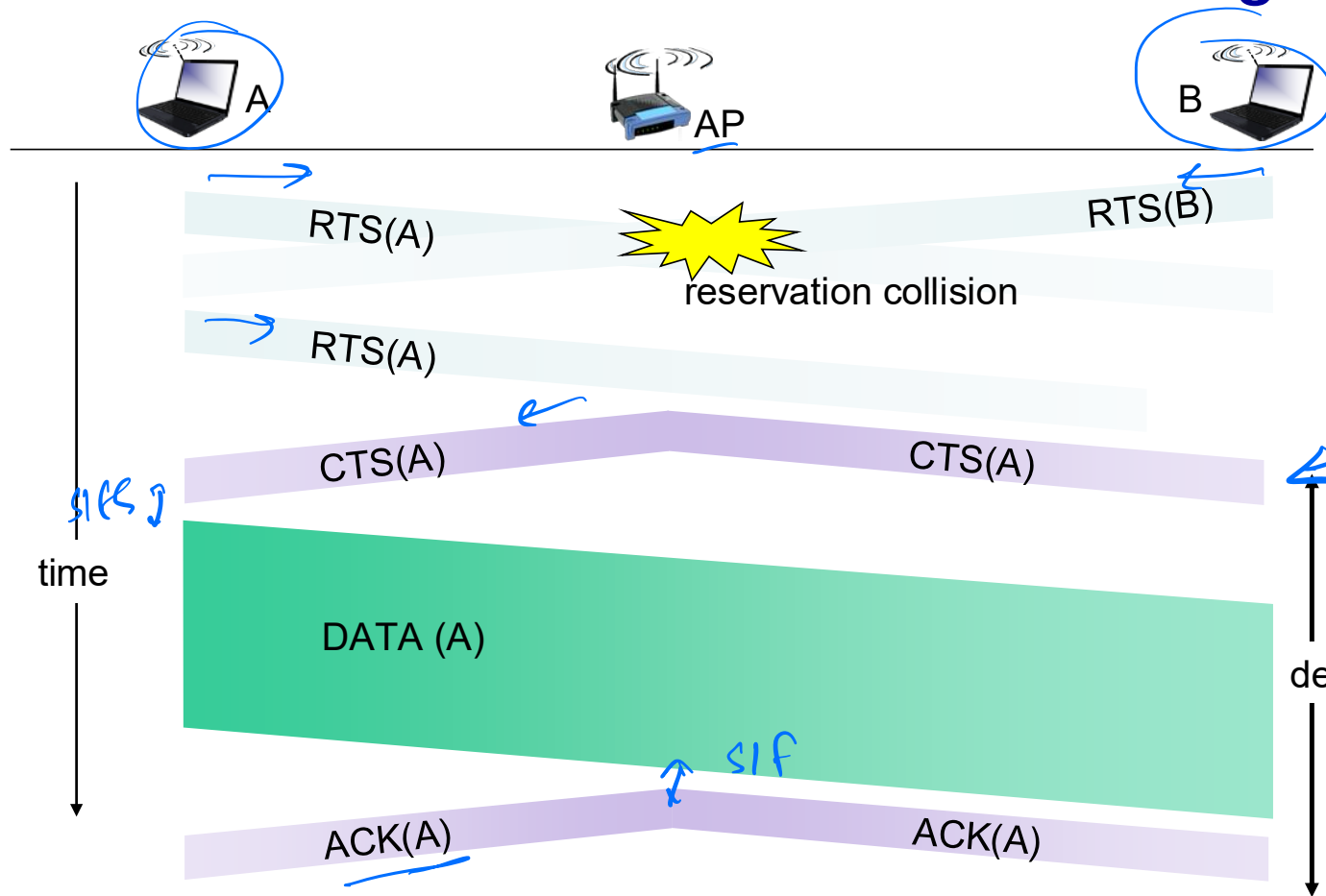
Avoiding collisions (hidden terminal problem)

idea: sender “reserves” channel use for data frames using small reservation packets

- sender first transmits *small* request-to-send (RTS) packet to BS using CSMA
 - RTSs may still collide with each other (but they're short)
- BS broadcasts clear-to-send (CTS) in response to RTS
- CTS heard by all nodes
 - sender transmits data frame
 - other stations defer transmissions

Collision Avoidance: RTS-CTS exchange

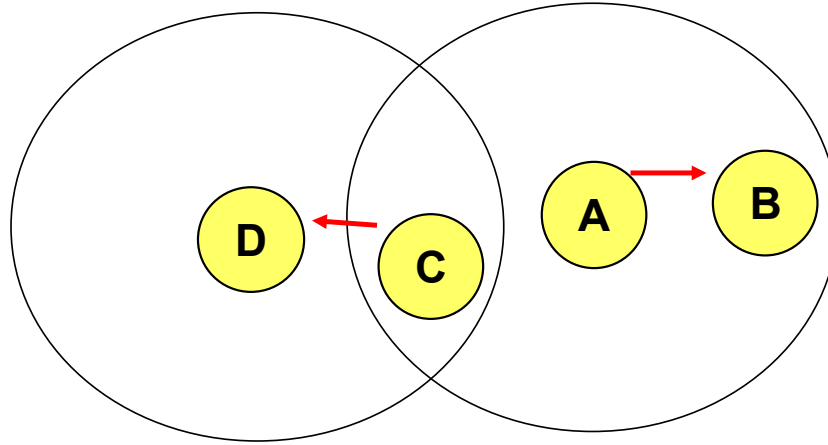
How it
solves
Hidden
Terminal
Problem?



Collision Avoidance: RTS-CTS exchange

- CSMA/CA alone does not effectively solve the hidden terminal problem because it relies on local sensing of the medium, which is insufficient when devices are out of range of each other.
- The RTS/CTS mechanism complements CSMA/CA and successfully addresses the hidden terminal problem by explicitly coordinating transmissions and preventing collisions caused by hidden nodes.

Exposed Terminal Problem

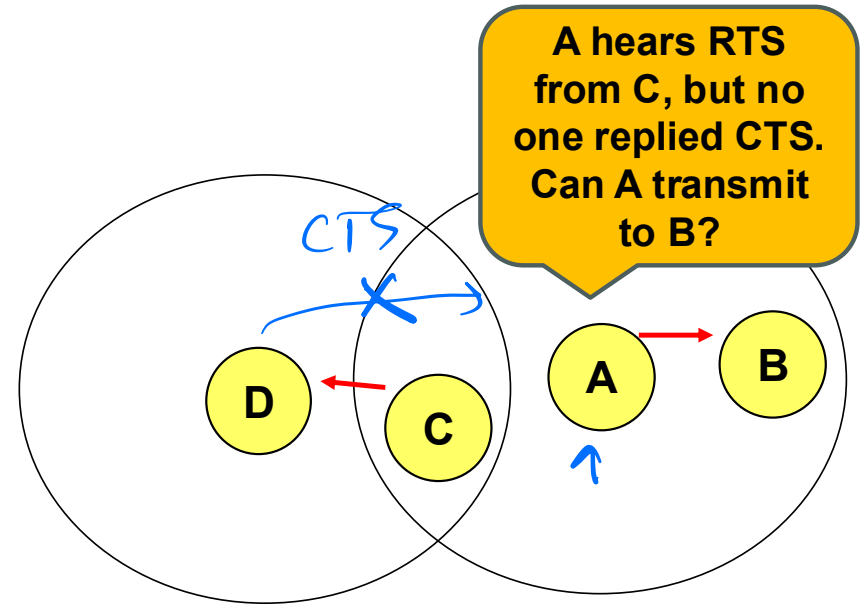


you listen to
Tx
not to Rx
but
collision happen
at Rx!

- Exposed terminals (A valid comm can not take place because the sender is exposed)
 - A starts sending to B.
 - C senses carrier, finds medium in use and has to wait for A->B to end.
 - D is outside the range of A, therefore waiting is not necessary.
 - A and C are “exposed” terminals i.e. A and C could communicate to their receivers at the same time (since their receivers are far apart), but because A and C are exposed to each other (i.e. can hear each other), one of them needlessly refrains from transmitting.
- A->B and C->D transmissions could have taken place in parallel without collisions, but CS function will only allow one of them

RTS/CTS to recover Exposed Terminal Problem

- Sense channel (CS)
- If busy
 - Freeze backoff timer
- Else
 - Send RTS
 - If CTS not received
 - Back-off to try again later
 - Else
 - Send Data
 - If ACK not received
 - Back-off to try again later
 - Next packet processing



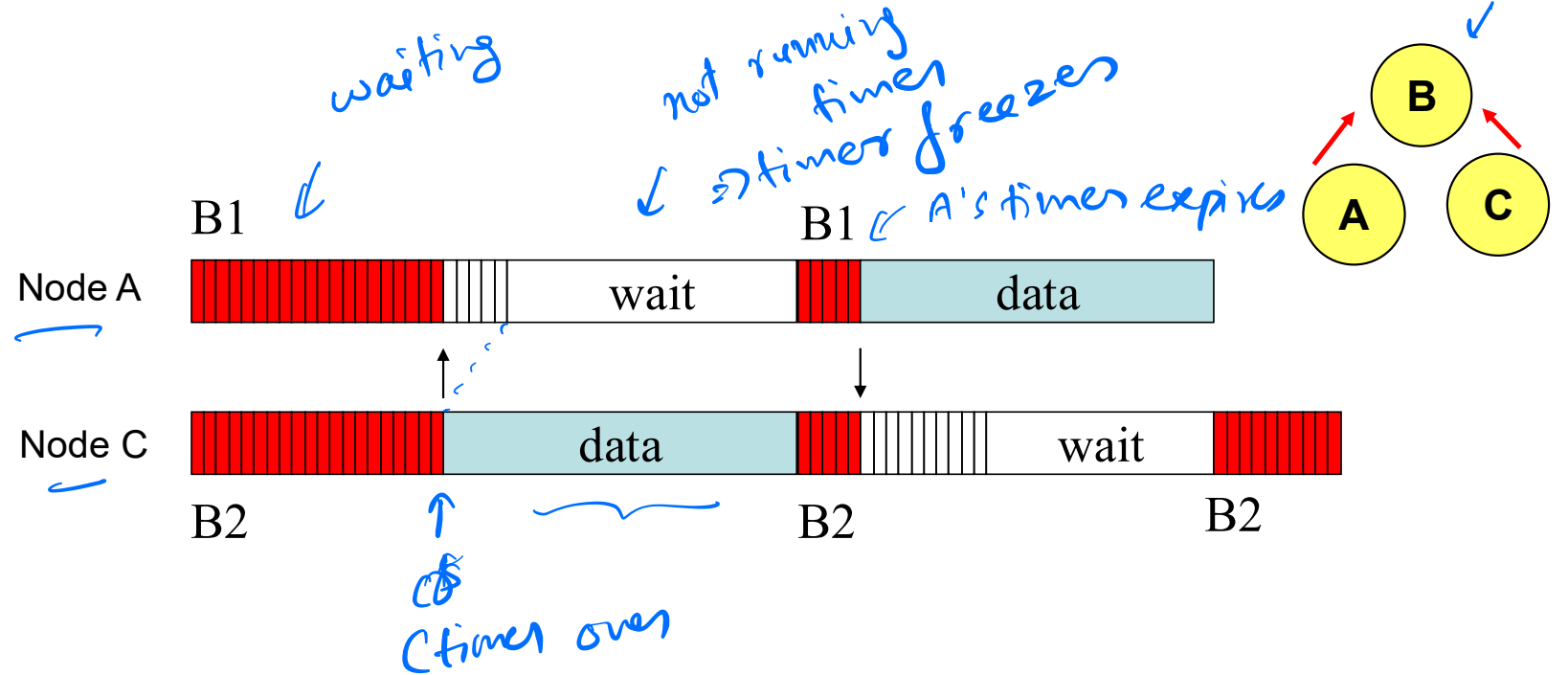
RTS/CTS helps in exposed node problem
When a node **hears an RTS from a neighboring node, but not the corresponding CTS**, that node can deduce that it is an exposed node and is permitted to transmit to other neighboring nodes.

CSMA/CA Algorithm (Contention Window)

- Maintain a value **CW** (Contention Window)
- If Busy,
 - Wait till channel is idle. Then choose a random number between 0 and CW and start a back-off timer for proportional amount of time.
 - If transmissions within back-off amount of time, freeze back-off timer and start it once channel becomes idle again.
- If Collisions (Control or Data)
 - Binary exponential increase (doubling) of CW

CSMA/CA Backoff Example

Rough example (no DIFS/SIFS)



B1 and B2 are backoff intervals
at nodes A and C

Backoff Interval

- The time spent counting down backoff intervals is a part of MAC overhead
- Choosing a *large cw* leads to large backoff intervals and can result in larger overhead
- Choosing a *small cw* leads to a larger number of collisions (when two nodes count down to 0 simultaneously)

Backoff Interval

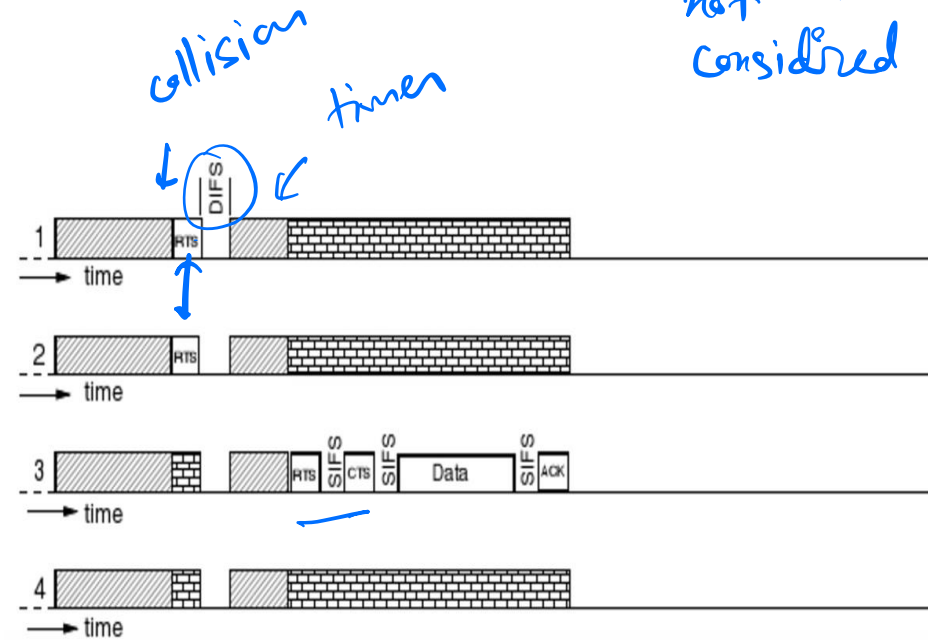
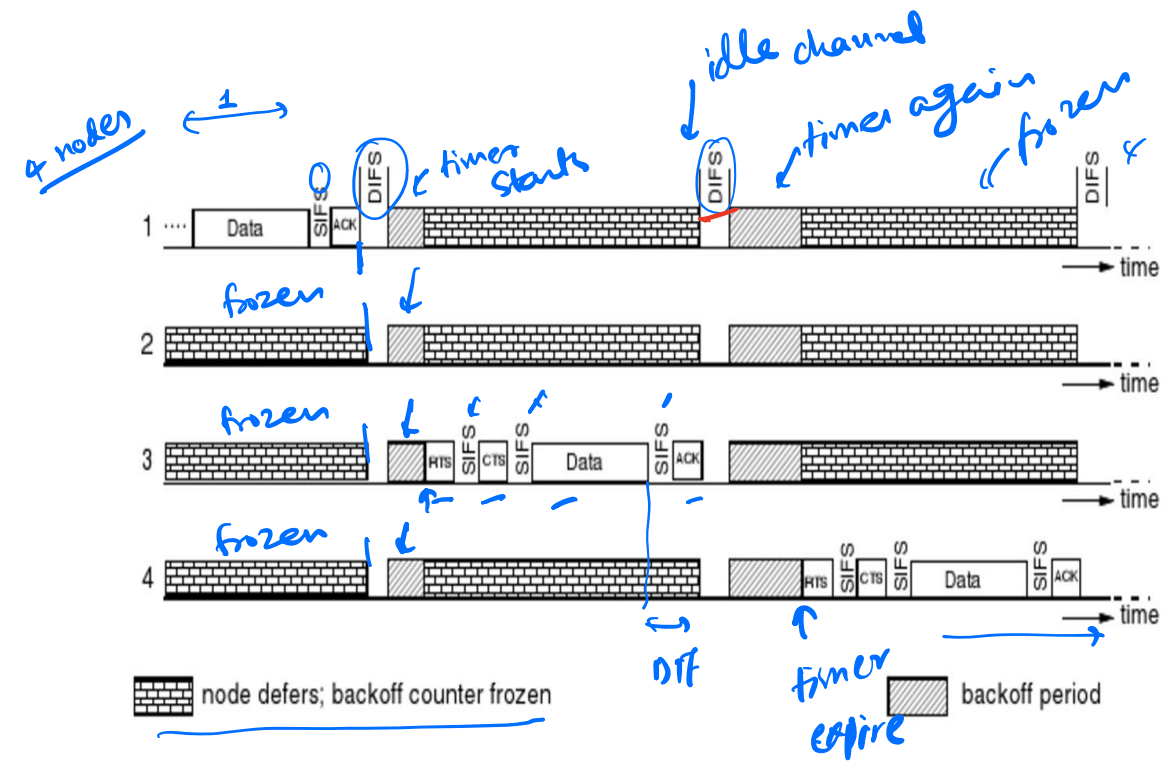
- Since the number of nodes attempting to transmit simultaneously may change with time, some mechanism to manage contention is needed
- IEEE 802.11 : contention window ***cw*** is chosen dynamically depending on collision occurrence

Binary Exponential Backoff in 802.11

- When a node fails to receive CTS in response to its RTS, it increases the contention window *or fails to receive ACK*
 - *cw* is doubled (up to an upper bound)
- When a node successfully completes a data transfer, it restores *cw* to *Cwmin* *Ack is received.*

Example scenario: 802.11 DCF exchange

(transmission delay $\times$ not considered)



Problem (similar to the HW2 problem)

In the 802.11 DCF timing diagram (see last slide), suppose the following frame spacing and transmission durations are given:

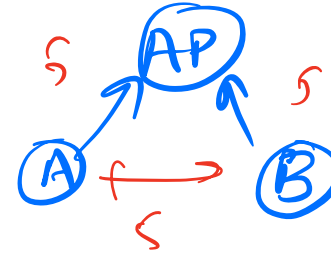
$$\text{DIFS} = 12$$

$$\text{SIFS} = 7$$

$$\text{ACK} = \text{RTS} = \text{CTS} = 15$$

$$\text{DATA} = 80$$

$$\text{Propagation delay} = 5$$



In case of a collision, nodes involved in the collision deterministically double their initial timer value.

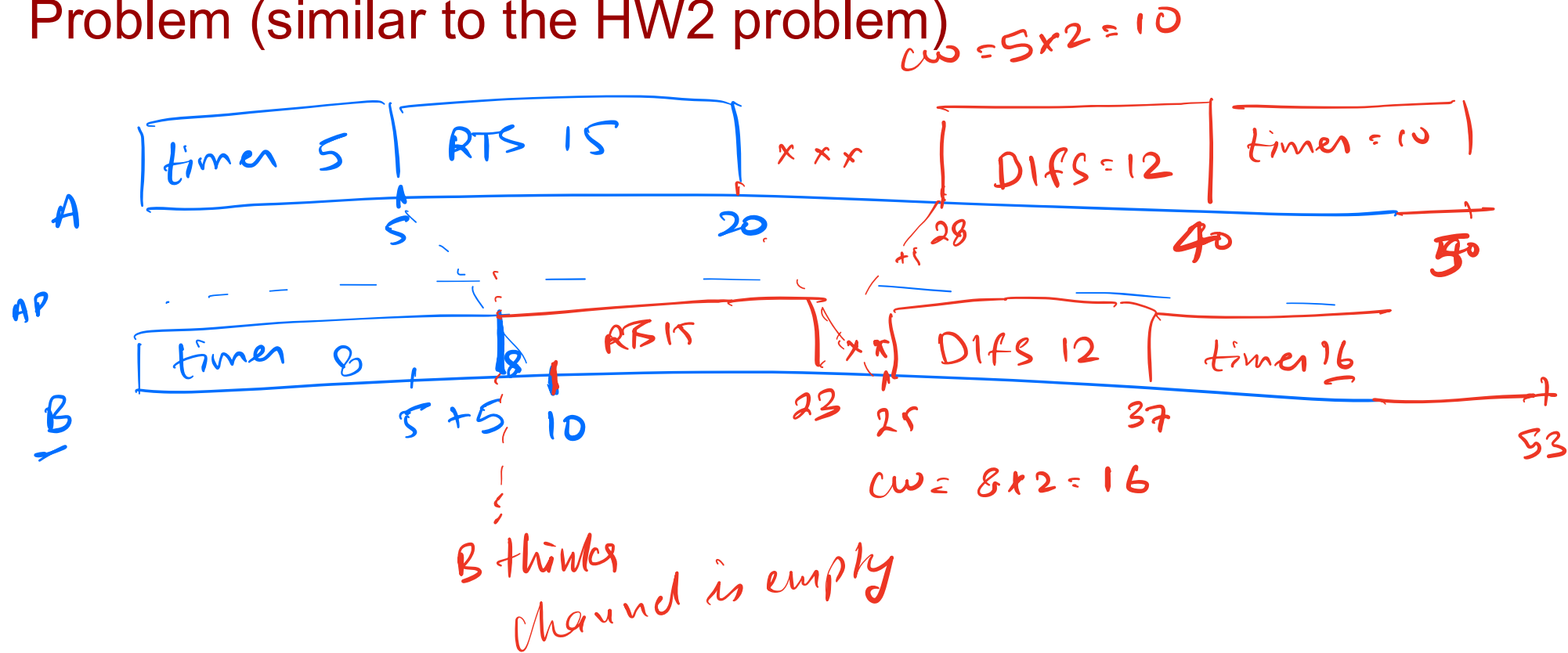
Assume there are two nodes, A and B. The initial condition is at $t=0$ where the channel has been idle for a long time, Node A timer = 5, Node B timer = 8, and each of the nodes has data to send, of duration DATA. The time until both nodes have completed received the data (i.e., completed receiving the latest of the two ACK packets) is _____?

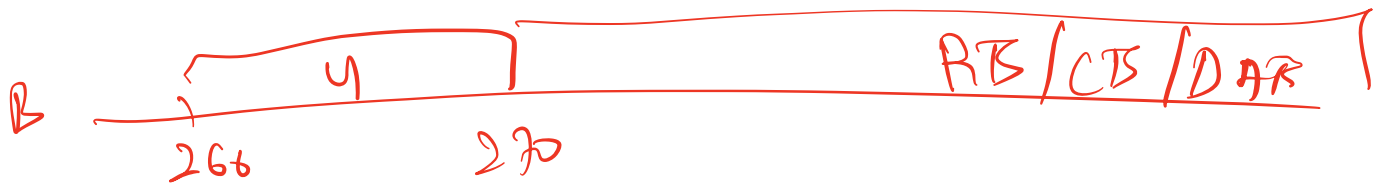
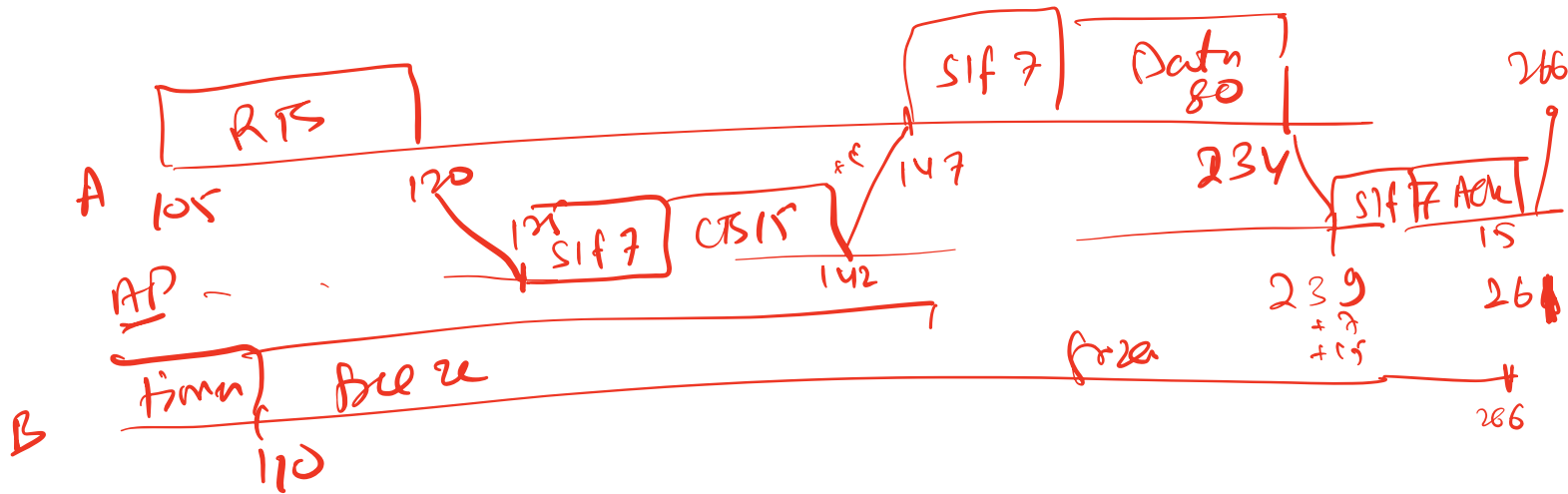
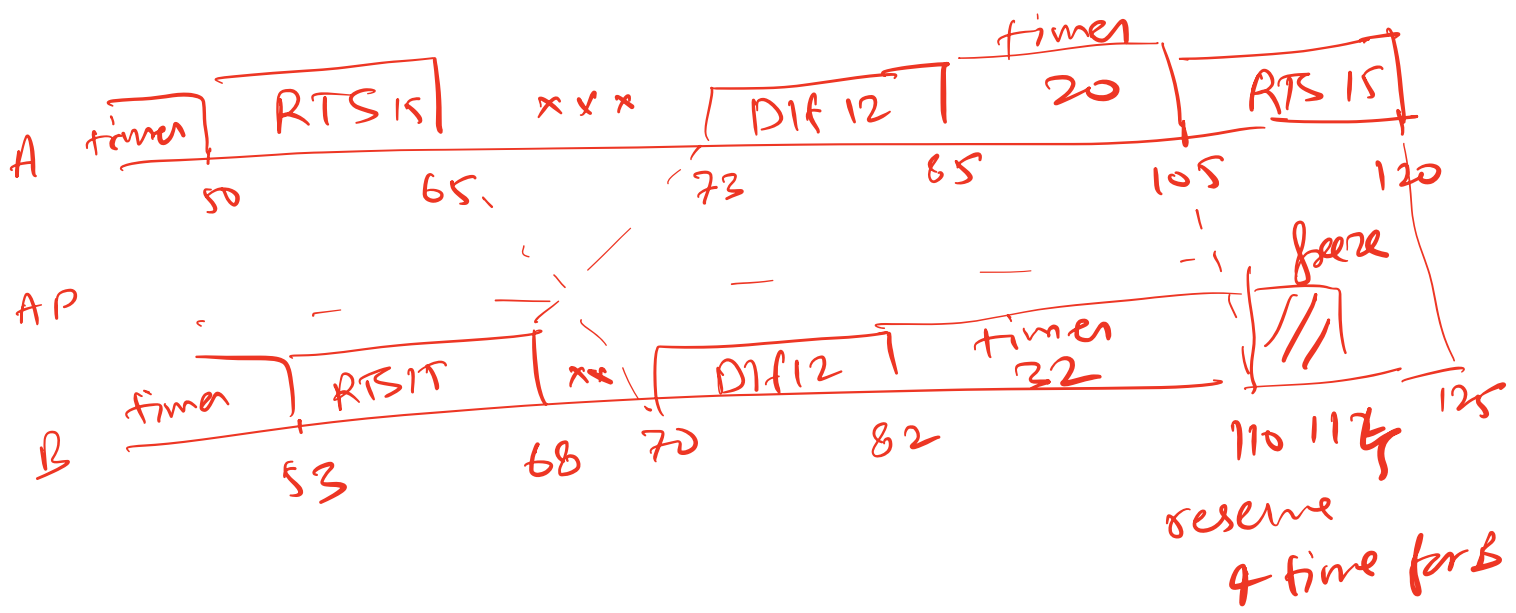
$$CW_A = 5$$

$$CW_B = 8$$

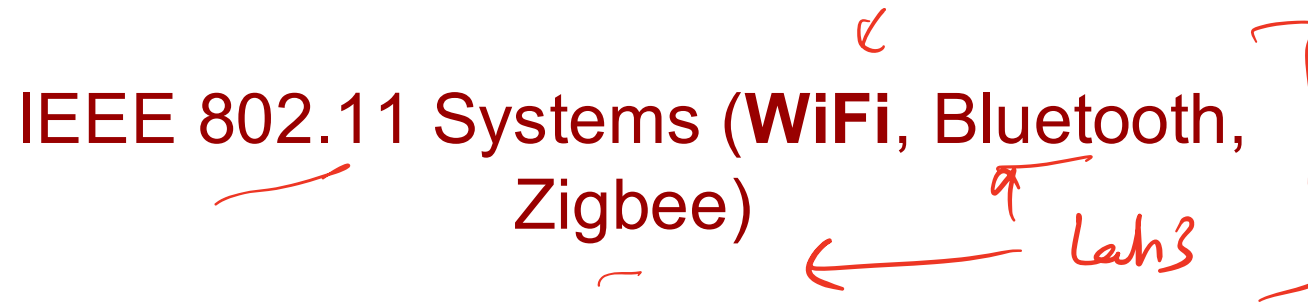
Assumption $\rightarrow$ no randomness

Problem (similar to the HW2 problem)





IEEE 802.11 Systems (**WiFi**, Bluetooth,
Zigbee)



IEEE 802.11

■ The 802.11 standard

- MAC and PHY functionality for wireless connectivity of fixed, portable and moving stations moving at pedestrian and vehicular speeds within a local area.

■ Specific features of the 802.11 standard include the following:

- Operates at either 2.4 GHz or 5 GHz unlicensed ISM bands *6 GHz*
- IEEE 802.11 a,g: Accommodation of transmission rates of 1, 2, 11, and 54 Mbps. *8x8 mimo*
- MIMO: n: 150Mbps, ac: 780Mbps *new standards*
- Support Multicast/broadcast services *⇒ better PHY technique*
- Network management services
- Registration and authentication services
- In certain modes, could support asynchronous and time-bounded delivery service

IEEE 802.11 modes of operation

■ Infrastructure Mode

- Each Mobile Station (MS) associates itself with an Access Point (AP)
- An AP+MS's associated with it are called a Basic Service Set (BSS)
- BSS's may (should? For handoff) overlap
 - An isolated BSS is called an Independent BSS
 - A set of AP's connected with a LAN are called an Extended BSS
- Modes of communication permitted are AP $\leftarrow \rightarrow$ MS

■ Ad-Hoc Mode

- MS $\leftarrow \rightarrow$ MS, i.e. no access point, CSMA used / CA

IEEE 802.11 Logical Architecture

- The logical architecture of the 802.11 standard that applies to each station consists of a single MAC and one of multiple PHYs

- PHYs

- Frequency hopping PHY
- Direct sequence PHY
- Infrared light PHY

← Bluetooth
← Zigbee

- MAC

- Point Coordination Function (PCF)
 - in infrastructure mode – centralized MAC
- Distributed Coordination Function (DCF)
 - in ad-hoc mode – distributed MAC

OFDM ← wifi

← equivalent to cellular

CSMA/CA

PHYs

- The original (~1997) 802.11 supported three options
 - FHSS, 2.4 GHz, 1 or 2 Mbps (depending on modulation scheme)
 - DSSS, 2.4 GHz, 1 or 2 Mbps
 - Infrared, 850-950-nm range, 1 or 2 Mbps

Newer standards:

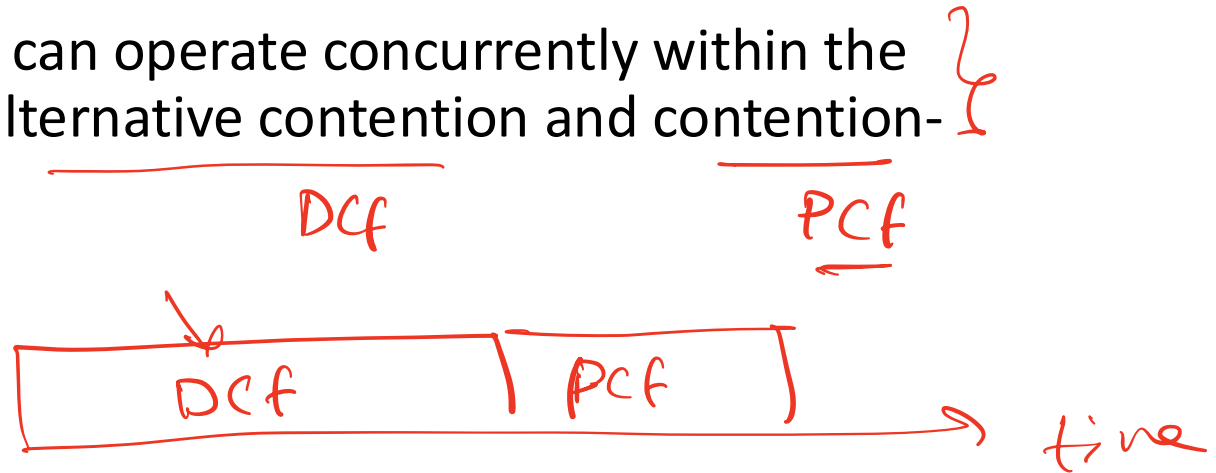
- 802.11b
 - Improvement to DSSS (2.4 GHz) to support rates of 5.5 and 11 Mbps (new modulation techniques)
- 802.11a
 - Uses OFDM in the 5 GHz band, rates up to 54 Mbps
- 802.11g
 - Aims at providing .11a speeds at 2.4 GHz band.
- 802.11n, ac, ax: utilize MIMO
 - Speeds 100 Mbps – 1 Gbps

→ wifi 6

wifi 5

IEEE 802.11 MAC

- DCF
 - CSMA/CA+RTS/CTS – A distributed contention based protocol
- PCF
 - Contention-free access protocol usable on infrastructure network configurations containing a controller (AP) called a point coordinator within the BSS
- Both the DCF and PCF can operate concurrently within the same BSS to provide alternative contention and contention-free periods



4G/5G Cellular Systems (3GPP)

UNITED
STATES
FREQUENCY
ALLOCATIONS

RADAR SERVICES COLOR LEGEND			
	RADIO METEOROLOGICAL	OTHER SATELLITE	RADAR ATTEMPTING
	AMBIENT/CLIMATE SERVICE SATELLITE	LAND SATELLITE	RADAR/INTERFEROMETRIC SATELLITE
	ENVIRONMENTAL SERVICE SATELLITE	LAND/COASTAL SATELLITE	RADAR/CLIMATE
	AGRICULTURE	MARITIME SATELLITE	RADAR/CLIMATE SATELLITE
	WATER/STRESS SATELLITE	MARITIME SATELLITE	RADAR/CLIMATE
	WATER/STRESS SATELLITE	MARITIME SATELLITE	RADAR/CLIMATE SATELLITE
	ENVIRONMENTAL SERVICE SATELLITE	METEOROLOGICAL	RADAR/CLIMATE
	WATER/STRESS SATELLITE	METEOROLOGICAL SATELLITE	RADAR/CLIMATE
	WATER	WATERS	RADAR/CLIMATE AND OTHER SERVICES
	OTHER SATELLITE	RADAR/CLIMATE SATELLITE	RADAR/CLIMATE AND OTHER SERVICES

ACTIVITY CODE

 INITIAL ACQUISITION

 INITIAL NON-REPEATED SEARCH

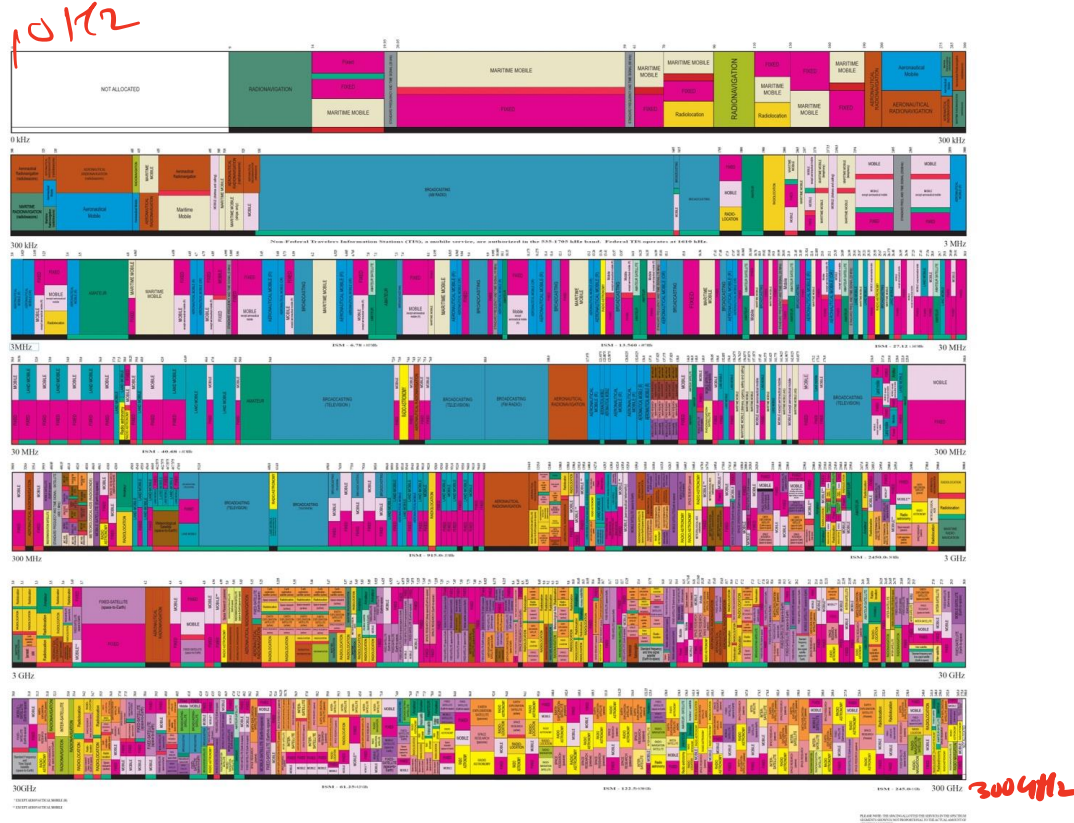
ALLOCATION USAGE DESIGNATION

SERVICE	EXAMPLE	DESCRIPTION
Primary	TTT2D	Capital Letters
Secondary	ttt2d	1st Capital with lower case letters

This chart is a graphic single-point-in-time portrayal of the Table of Frequency Allocations used by the FCC and ETSI. As such, it may not completely reflect all aspects, or details and recent change made to the Table of Frequency Allocations. Therefore, for complete information, users should consult the Table to determine the current state of FCC allocations.



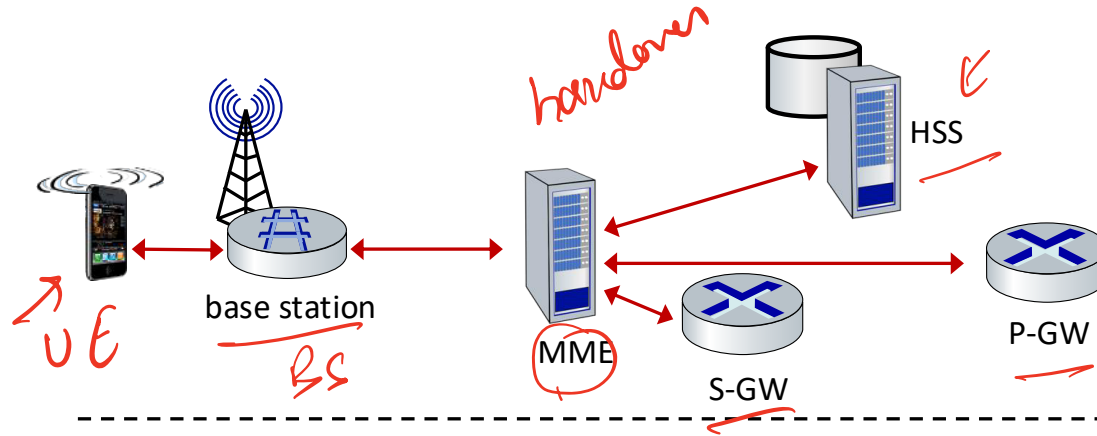
U.S. DEPARTMENT OF COMMERCE
National Telecommunications and Information Administration
Office of Spectrum Management
JANUARY 2016



<https://transition.fcc.gov/oet/spectrum/table/fcctable.pdf>

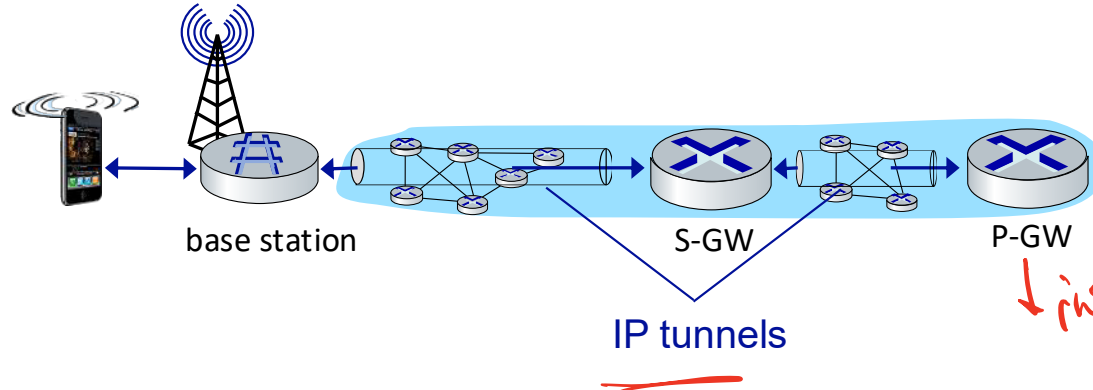
...but spectrum is not used efficiently at all!

LTE: data plane control plane separation



control plane

- new protocols for mobility management, security, authentication (later)



data plane

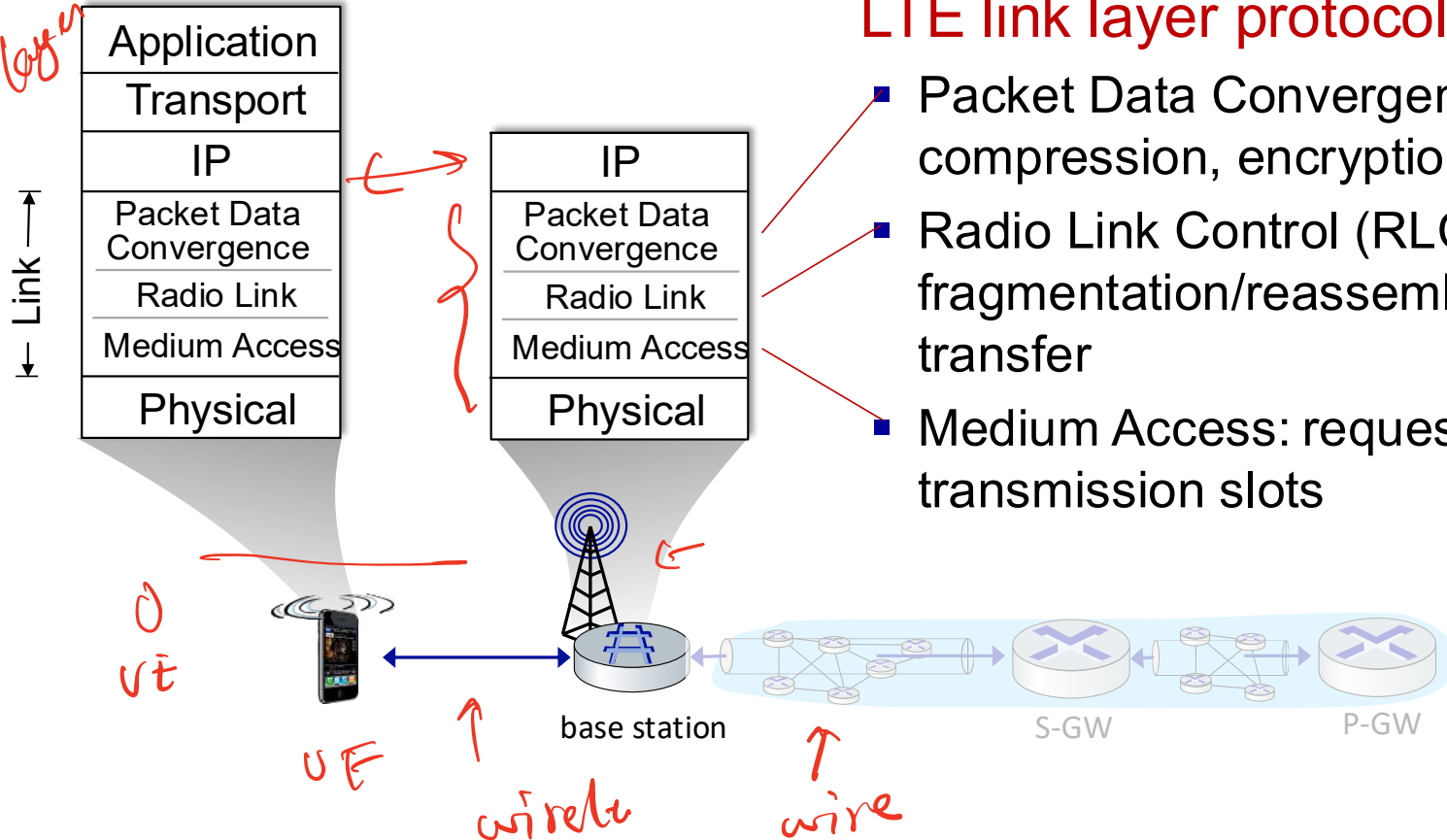
- new protocols at link, physical layers
- extensive use of tunneling to facilitate mobility

Serving Gateway (S-GW), PDN Gateway (P-GW)
HSS: Home Subscriber Service; MME: Mobility Management Entity

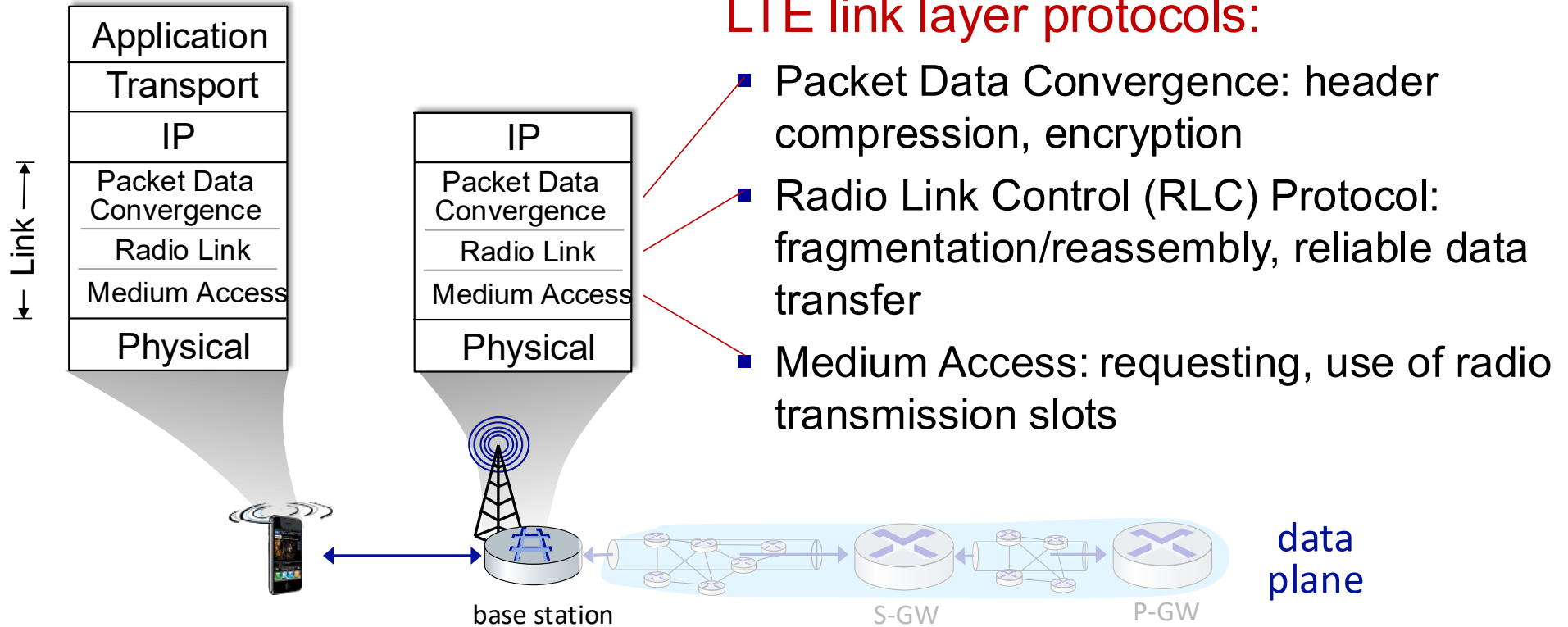
LTE data plane protocol stack: first hop

LTE link layer protocols:

- Packet Data Convergence: header compression, encryption
- Radio Link Control (RLC) Protocol: fragmentation/reassembly, reliable data transfer
- Medium Access: requesting, use of radio transmission slots



LTE data plane protocol stack: first hop

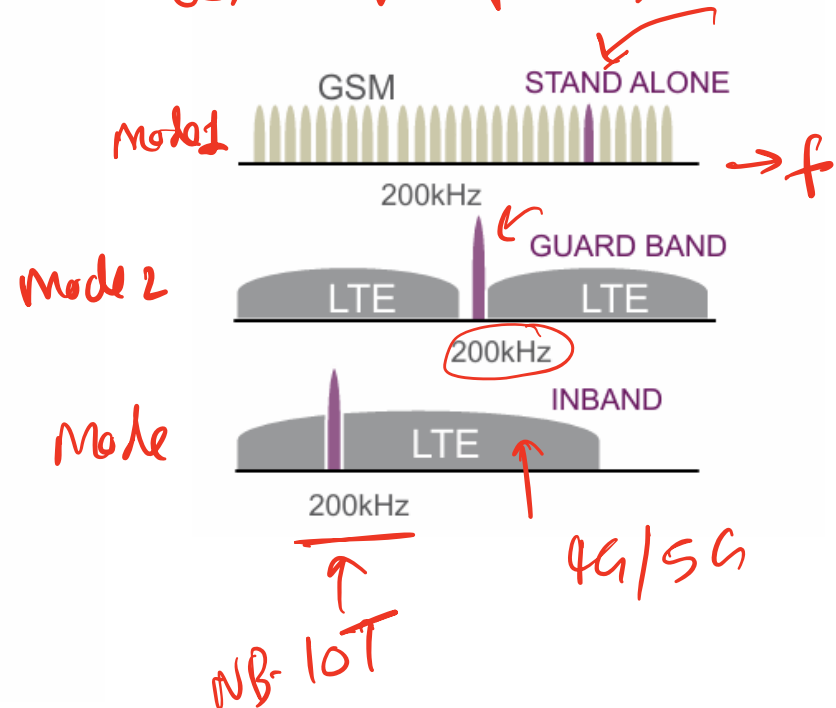


4G LTE cellular networks and NB-IoT

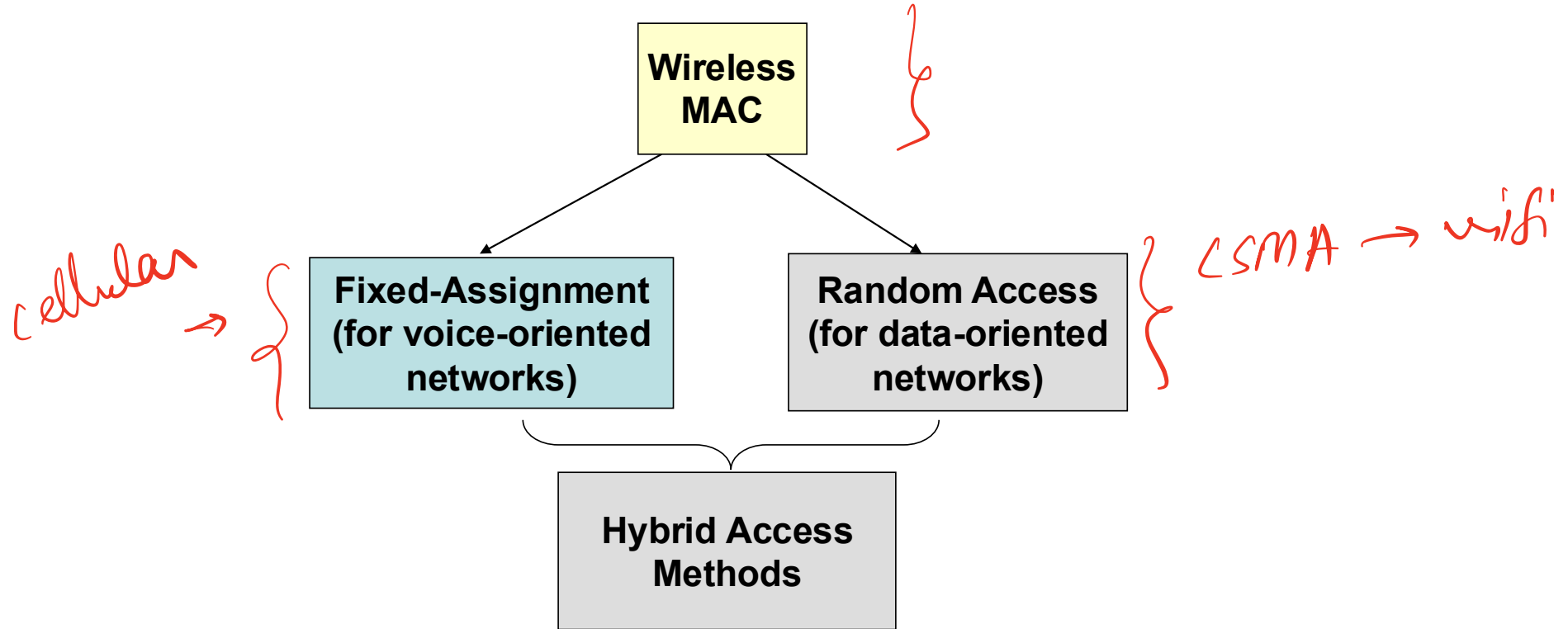
- Targeting implementation in an existing 3GPP network
- Applicable in any 3GPP defined (licensed) frequency band – standardization in release 13
- Three deployment modes
- Processing along with wideband LTE carriers implying OFDM secured orthogonality and common resource utilization
- Maximum user rates 30/60 (DL/UL) kbps

Narrow-band
IoT

Exist on top of 4G/5G
with low power connection

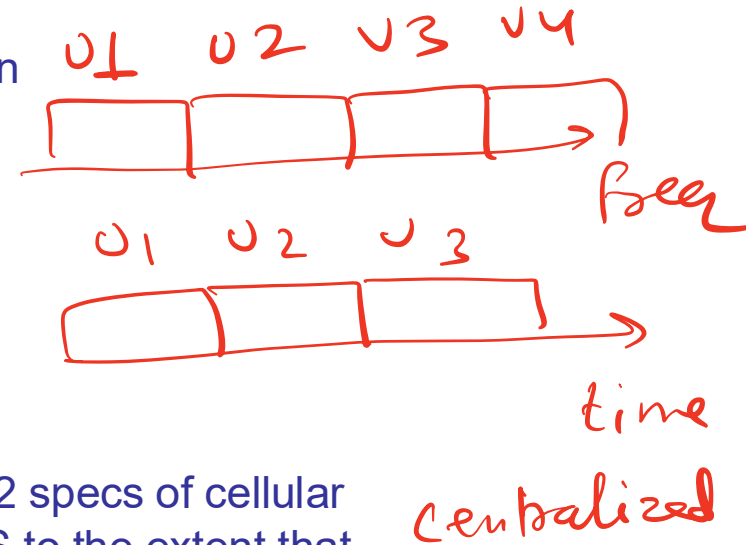


Wireless MAC Classification



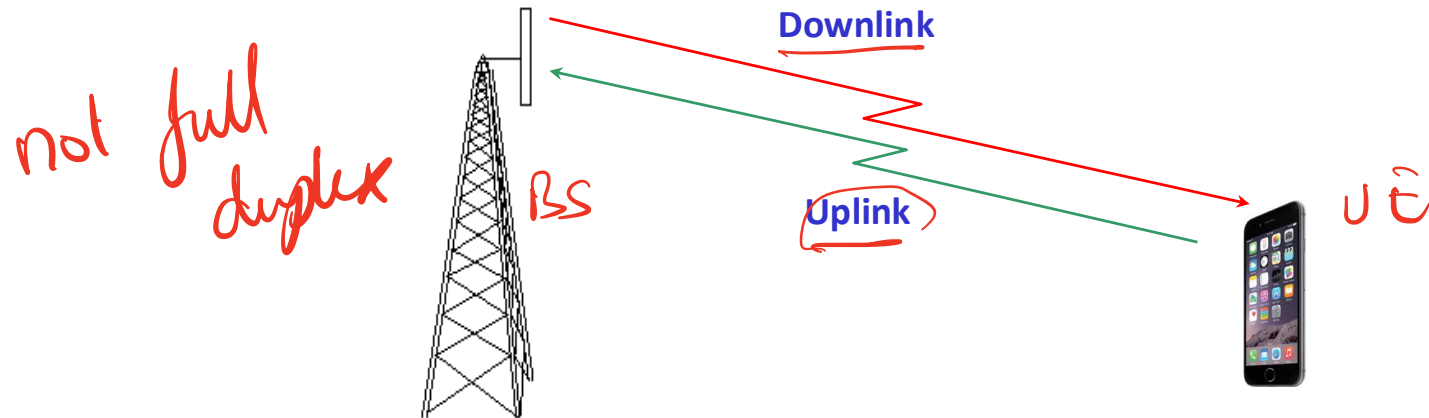
Fixed Assignment MAC

- Used by all existing wide area wireless networks (e.g. cellular)
- Note: Some cellular networks use random access methods in addition (e.g. GSM)
- Fixed allocation of resources for the duration of a connection
- Three basic fixed-assignment multiple-access methods:
 - FDMA: Frequency Division Multiple Access
 - TDMA: Time Division MA
 - CDMA: Code Division MA ← 3G
- Although channel access method is only a part of the layer 2 specs of cellular networks, choice of MAC method impacts capacity and QoS to the extent that we refer to various voice-oriented wireless systems by their channel access method



Forward vs. Reverse Channels

- Differentiation between the forward and reverse channels:
 - Forward: (downlink) between the base station and mobile terminal
 - Reverse: (uplink) between the mobile terminal and the base station

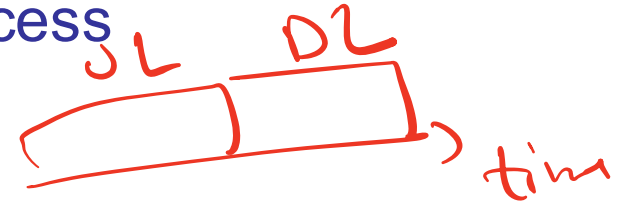


Duplexing

An important design parameter related to the access mechanism is the method used for “duplexing”

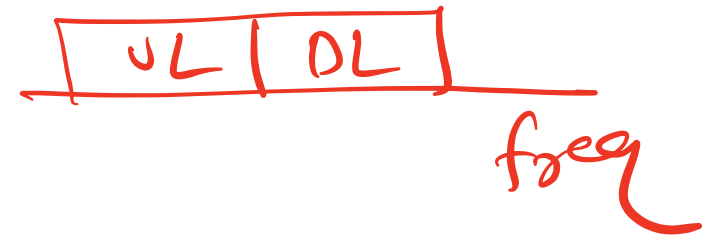
- TDD:

- Forward and reverse channels use the same frequency band
- Forward and reverse channels use alternating timeslots



- FDD:

- Forward and reverse channels use different and sufficiently separated carrier frequencies.



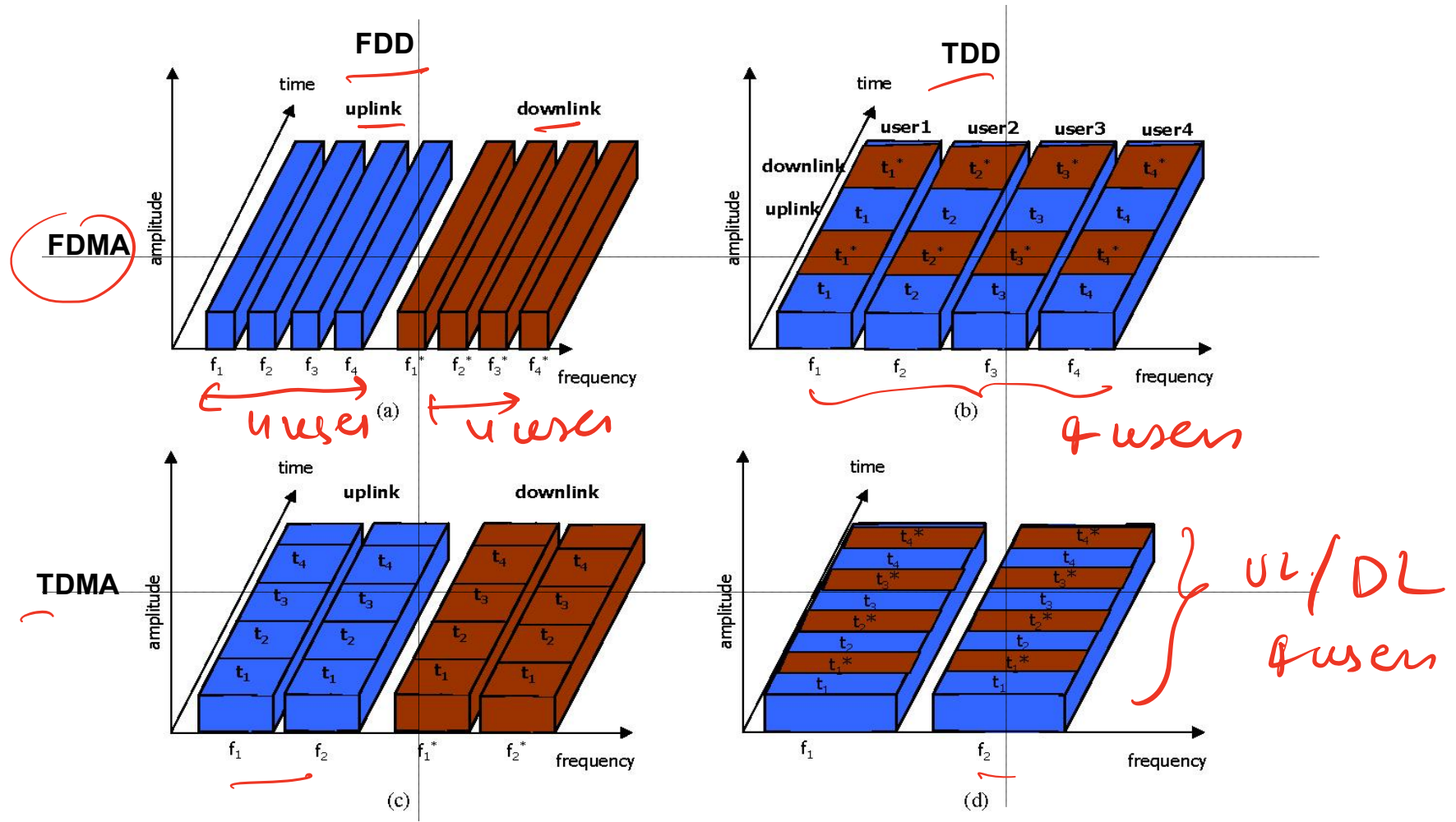


Figure 1: (a) FDMA/FDD (b) FDMA/TDD (c) TDMA/FDD with multiple carriers (d) TDMA/TDD with multiple carriers

Practical wideband digital modulation

- DSSS (Direct Sequence Spread spectrum)
 - ✓ Used in ZigBee (802.15.4), WiFi (802.11b), WCDMA, etc.
- OFDM (Orthogonal frequency division multiplexing)
 - ✓ Used in 802.11a/g/n/ac/ad..., 4G LTE, 5G NR, etc.

CDMA *↳ codes*

■ Recall:

- FDMA: each user occupies a frequency-slot throughout the communication session
- TDMA: each user occupies a time-slot throughout the communication session

■ CDMA

- Multiple users use the same time and frequency slots at the same time!
- A user is differentiated by a code that acts as the key to identify that user
- Orthogonal codes are assigned such that when they are used at the same time in the same band, a receiver knowing the code of the particular user can detect that user signal among all simultaneously received signals from other users
- Here also, two versions exist:
 - CDMA/FDD: forward and reverse channels use different carriers
 - CDMA/TDD: forward and reverse channels use same carrier

CDMA Illustration

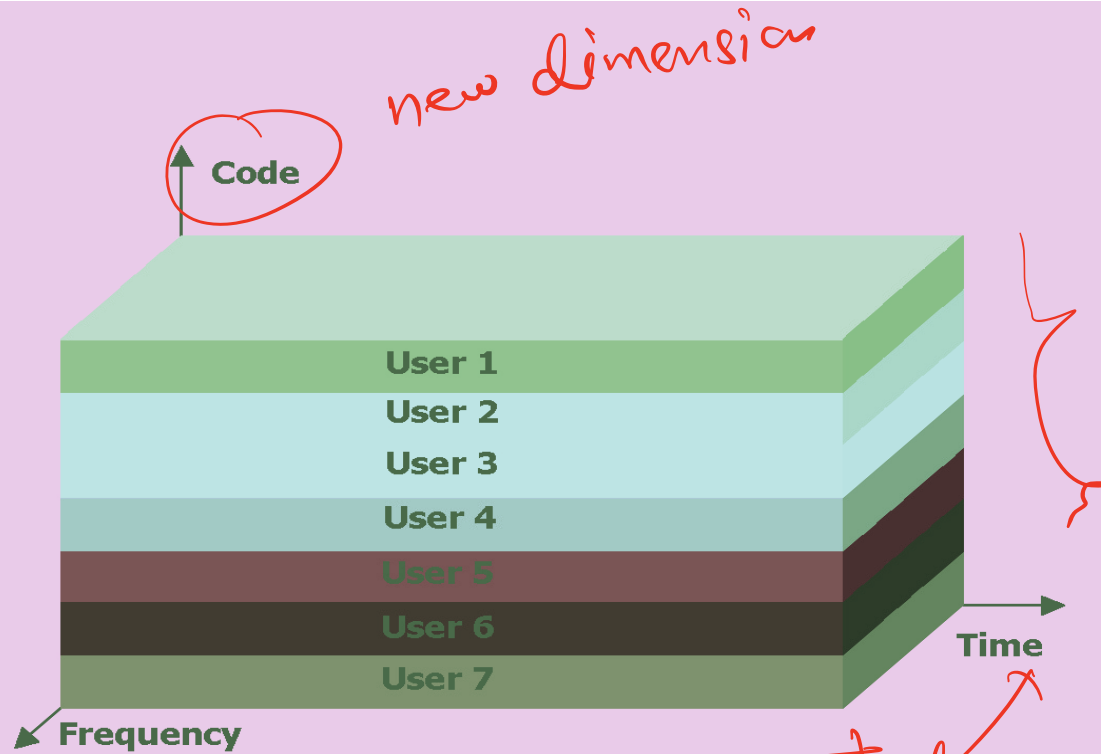


Figure 6: Simple illustration of CDMA

CDMA/FDD and CDMA/TDD Illustration

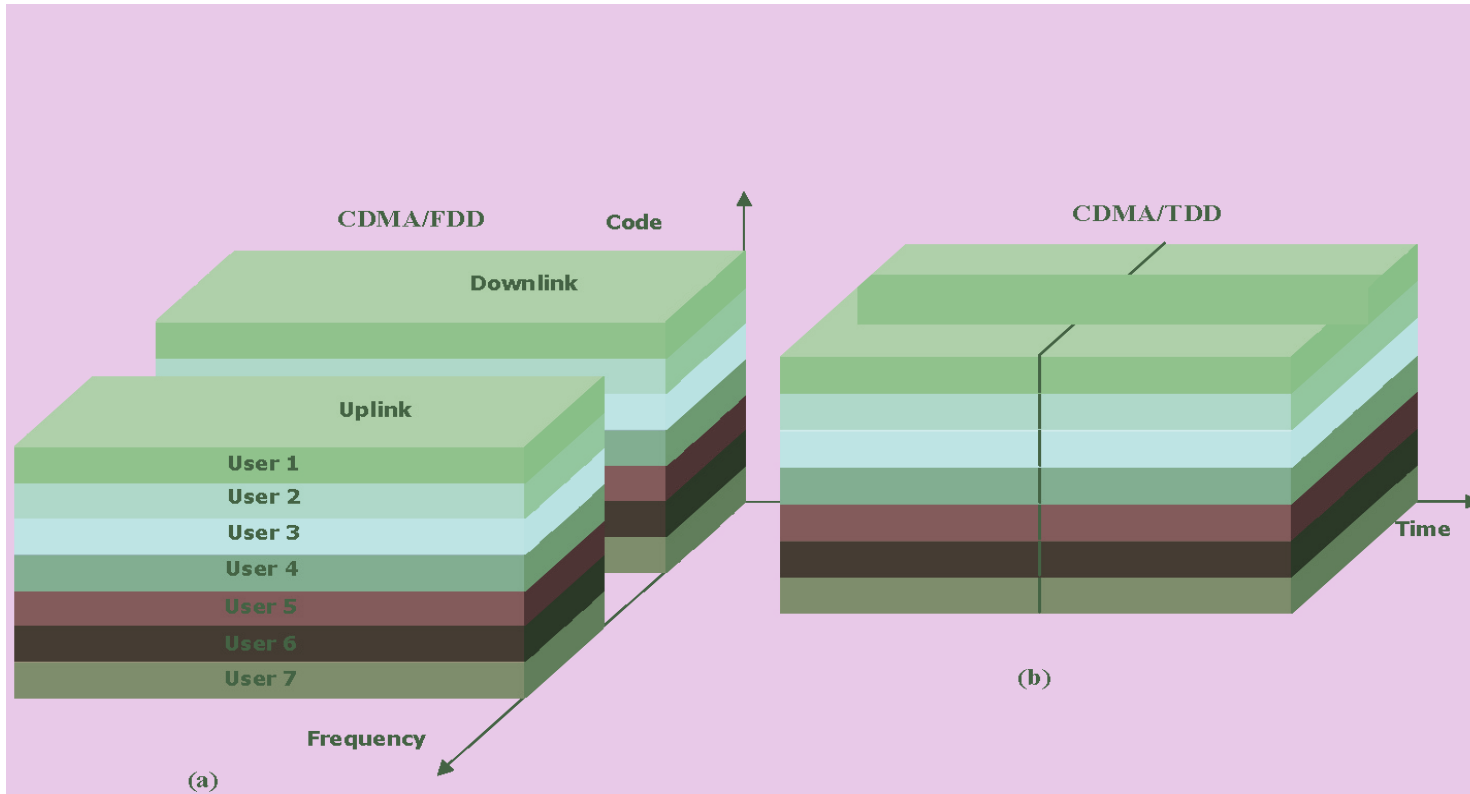
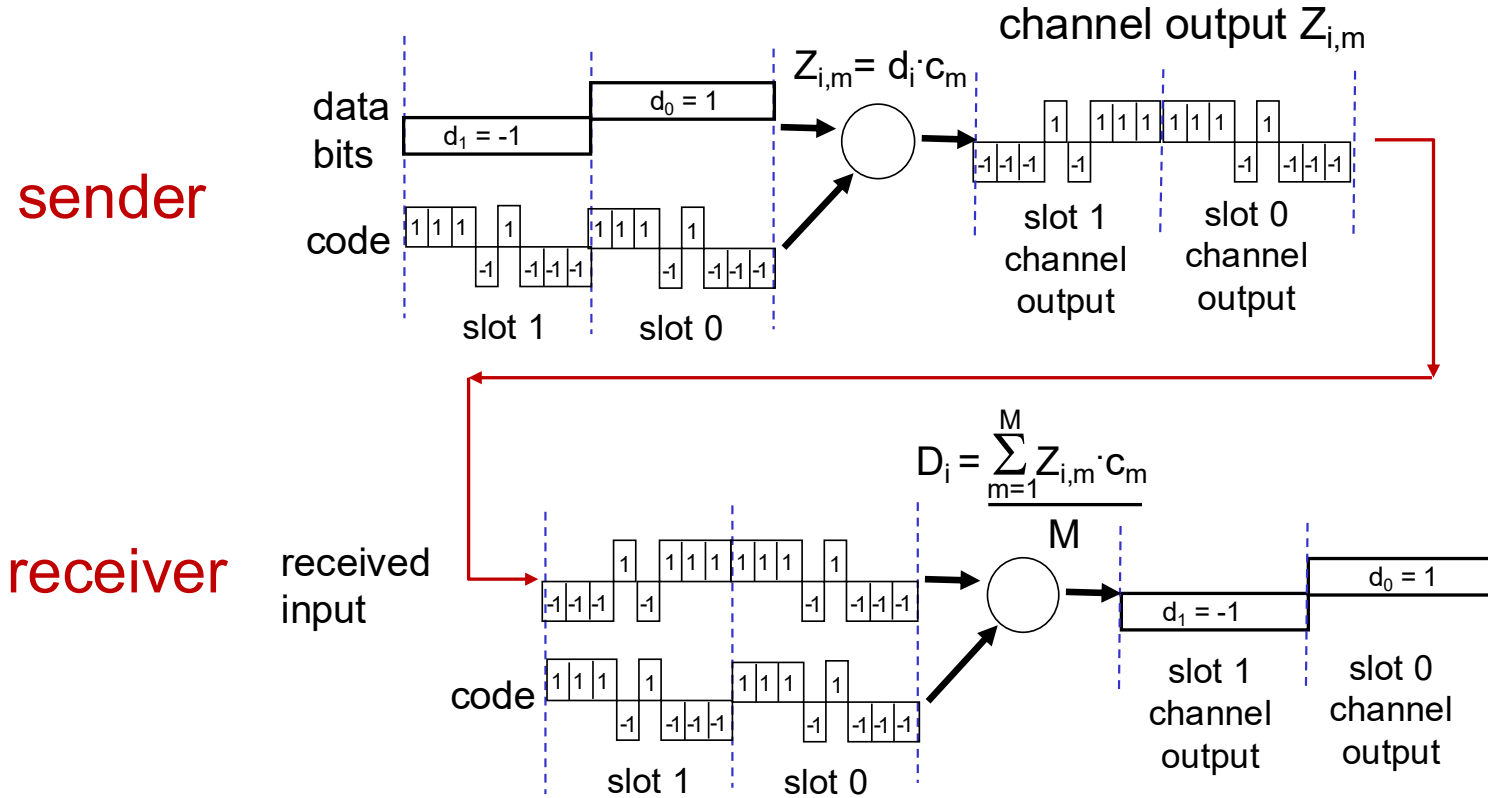


Figure 7: (a) Frequency and (b) time division duplexing with CDMA

Code Division Multiple Access (CDMA)

- unique “code” assigned to each user; i.e., code set partitioning
 - all users share same frequency, but each user has own “chipping” sequence (i.e., code) to encode data
 - allows multiple users to “coexist” and transmit simultaneously with minimal interference (if codes are “orthogonal”)
- **encoding:** inner product: (original data) $\times$ (chipping sequence)
- **decoding:** summed inner-product: (encoded data) $\times$ (chipping sequence)

CDMA encode/decode

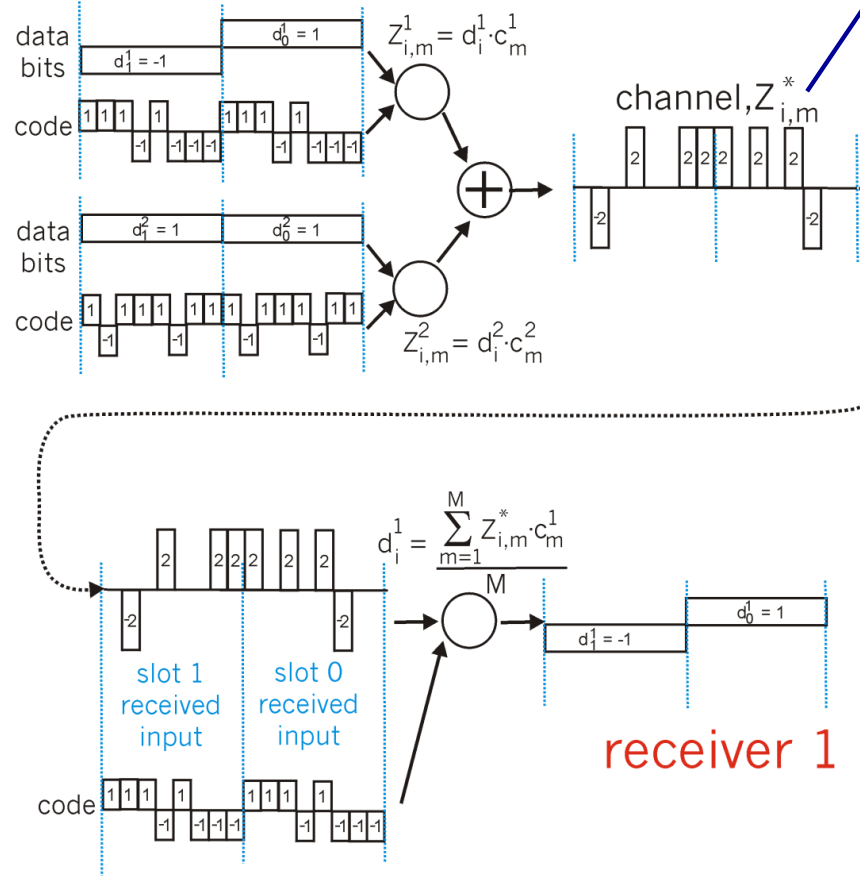


... but this isn't really useful yet!

CDMA: two-sender interference

Sender 1

Sender 2



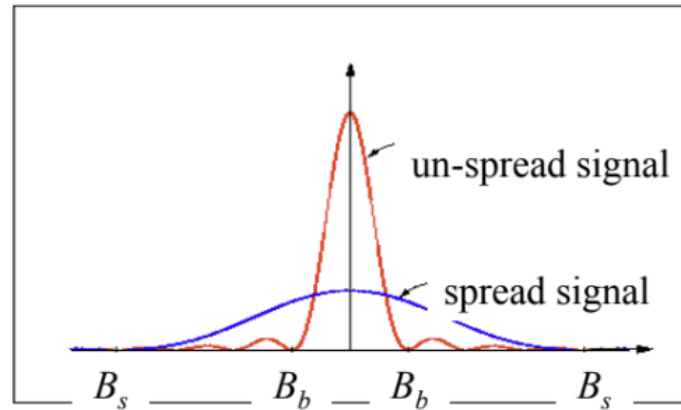
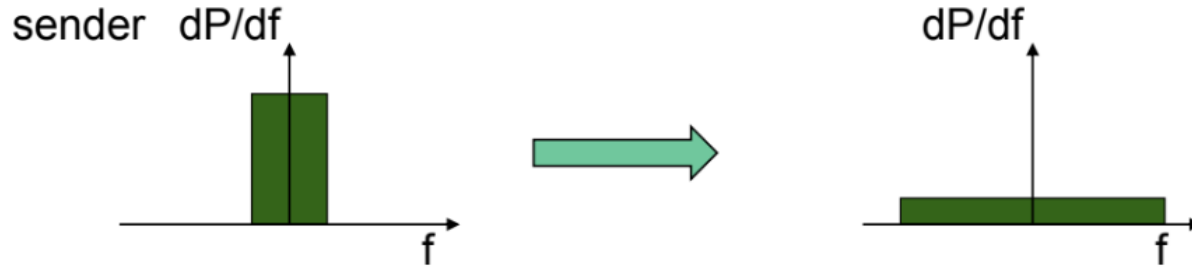
channel sums together transmissions by sender 1 and 2

using same code as sender 1, receiver recovers sender 1's original data from summed channel data!

receiver 1

... now *that's* useful!

Effect of spreading: frequency-domain perspective



B_s : num. of bits in the chip * B_b

Problem (similar to the HW2 problem)

A CDMA system operates with:

- **User Data Rate:** 200 kbps
- **Chipping Sequence Length:** 10
- **Total Users:** 6

Question: What is the **spectral efficiency** of the system in bits/sec/Hz?